

# **Barangay Incident Reporting and Management System with GIS Mapping**

A Thesis Project presented to the  
College of Computer Studies of  
Pamantasan ng Lungsod ng Pasig

In partial fulfillment of the requirements for the degree of  
Bachelor of Science in Information Technology

by:

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## TABLE OF CONTENTS



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## **ABSTRACT**

**Title: Barangay Incident Reporting and Management System with GIS Mapping**

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**Degree: Bachelor of Science in Information Technology**

**Year: 2026**

**Project Description:**

**This study presents the development of a web-based Barangay Incident Reporting and Management System integrated with Geographic Information System (GIS) mapping to improve the efficiency of reporting, tracking, and responding to community incidents. Traditional methods of incident reporting in barangays often rely on manual logs, verbal reports, and delayed documentation, leading to inefficiencies and lack of accurate records.**

**The proposed system allows residents to report incidents such as theft, accidents, or emergencies through an online platform. Reports are automatically tagged with location data and visualized using GIS mapping, enabling barangay officials to**



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quickly identify incident hotspots and respond accordingly. The system also includes features such as case tracking, report generation, and notification alerts. The system was developed using Agile methodology and evaluated based on ISO 25010 software quality standards, including functionality, usability, reliability, performance efficiency, security, maintainability, and portability. Evaluation results show that the system enhances response time, improves record management, and supports data-driven decision-making for barangay officials.



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## **CHAPTER 1: INTRODUCTION**

### **I. Introduction**

Barangays serve as the primary administrative units responsible for maintaining peace and order within communities. Efficient incident reporting and management are essential to ensure public safety and timely response to emergencies. However, many barangays still rely on traditional methods such as handwritten reports and verbal communication, which can lead to delayed responses, incomplete records, and poor data organization.

With the advancement of web technologies and GIS mapping, there is an opportunity to modernize incident reporting systems. Integrating location-based data with reporting platforms allows for better visualization and analysis of incidents, helping officials make informed decisions.

This study proposes a Barangay Incident Reporting and Management System with GIS mapping to streamline reporting processes, improve response efficiency, and enhance data accessibility.

### **II. Significance of the Study**

This system will benefit the following:

Residents – Easier and faster way to report incidents

Barangay Officials – Improved monitoring and response to community issues



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Law Enforcement – Access to organized and location-based incident data

Community – Enhanced safety and transparency

### **III. Statement of the Problem**

#### **General Problem**

How can a GIS-integrated system improve the efficiency of barangay incident reporting and management?

#### **Specific Problems**

1. How to provide a convenient platform for residents to report incidents?
2. How to accurately capture and visualize incident locations?
3. How to improve response time and coordination among officials?
4. How to maintain organized and secure incident records?

### **IV. Scope and Limitations**

#### **Scope**

- Online incident reporting system
- GIS-based location mapping
- Case tracking and status updates
- User roles: Admin, Barangay Staff, Resident

#### **Limitations**



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- Requires internet access for reporting and monitoring
  - Accuracy depends on user-provided location data
  - Limited to barangay-level implementation

### **OPERATIONAL DEFINITION OF TERMS**

**Administrator** – The personnel responsible for managing system settings, users, and reports.

**Barangay** – The smallest administrative division in the Philippines responsible for local governance.

**Case Tracking** – The process of monitoring the progress and status of reported incidents.

**Database** – A structured system for storing incident reports, user data, and location information.

**Frontend** – The user interface where residents and officials interact with the system.

**Geographic Information System (GIS)** – A system used to capture, store, and visualize geographic data.

**Incident Report** – A documented record of an event such as a crime, accident, or emergency.

**Location Data** – Geographic coordinates used to identify where an incident occurred.

**Notification System** – A feature that alerts officials about new or urgent reports.



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**Backend** – The server-side component that processes data and manages system logic.

**Role-Based Access Control (RBAC)** – A system that restricts access based on user roles.

**Security (ISO 25010)** – Measures implemented to protect sensitive incident data.

**Usability (ISO 25010)** – The ease with which users can interact with the system.